

SMART INDIA HACKATHON 2026

IDEA SUBMISSION

VALIDRA

Evidence-first AI compliance engine for packaged commodities

Problem Statement ID	26034
Problem Statement Title	Software System to check compliance of Packaged Commodities under Legal Metrology (Packaged Commodities) Rules, 2011 by scanning products, images and labels
Theme	Miscellaneous
PS Category	Software
Team Name	VisionMinds



THE PROBLEM

- **Mandatory declarations are critical:** Packaged commodities must display MRP, net quantity, manufacturer/packer details, dates and consumer-care information.
- **Manual inspection does not scale:** Checking millions of packaged products across retail and e-commerce is time-consuming and difficult to perform consistently.
- **Violations are easy to miss:** Missing declarations, incorrect MRP format, undersized text and improper placement require detailed visual verification.
- **No evidence-linked workflow:** Traditional inspection often separates the observed label evidence from the rule/provision used to justify the finding.

OUR SOLUTION - VALIDRA

AI extracts → Rule Engine validates → Evidence-backed report

- 1 **SCAN:** Upload or capture product label images for inspection.
- 2 **EXTRACT:** Image preprocessing + OCR identify required declarations and their location.
- 3 **VALIDATE:** A versioned rule engine checks presence, value, format, placement and font-size requirements against the applicable Legal Metrology rules.
- 4 **CLASSIFY:** Each declaration is classified as COMPLIANT, NON-COMPLIANT or NEEDS REVIEW based on validation and confidence.
- 5 **REPORT:** Generate a compliance report containing detected values, evidence crops, violations, confidence and applicable rule references.
- 6 **REVIEW:** Officers can review flagged cases and access searchable inspection history..

UNIQUENESS & INNOVATION

- **Evidence-first compliance** — Every decision is linked to the extracted value, image evidence and applicable rule.
- **Rule-based validation** — A dedicated rule engine checks presence, correctness, format, placement and font-size requirements instead of relying on an LLM for compliance decisions.
- **Confidence-aware pipeline** — Confidence is carried from OCR → extraction → validation; uncertain cases are automatically marked NEEDS REVIEW.
- **Multi-parameter checking:** checks declarations across value, format, placement, readability and font size, not just text presence.
- **Evidence-grade audit trail** — Original images, bounding boxes, extracted values, validation results and inspection records are preserved for traceability and review..

COMPLIANCE CHECKS PERFORMED



AI-ASSISTED

Automated label extraction with OCR + computer vision

RULE-DRIVEN

Compliance decisions based on predefined Legal Metrology rules

EVIDENCE-BACKED

Every finding includes extracted values and visual evidence

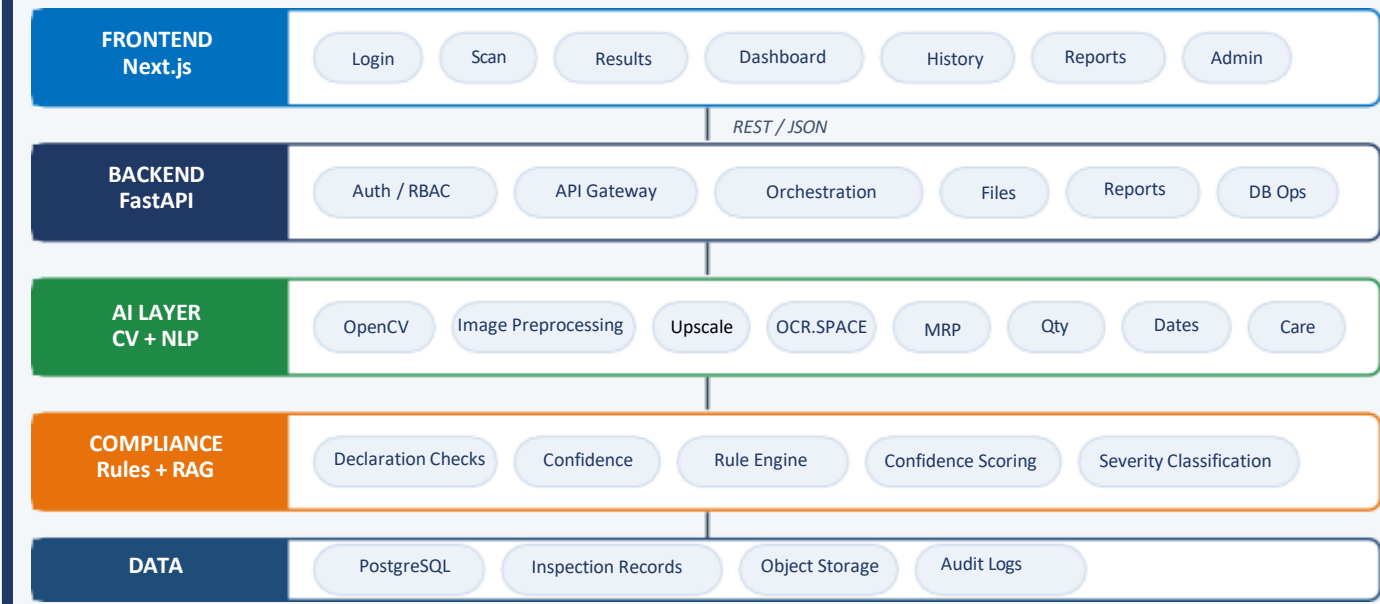
OFFICER-READY

Clear compliance status, flagged issues and downloadable reports

PROCESSING PIPELINE



ARCHITECTURE OVERVIEW



TECHNOLOGY STACK

Frontend	Next.js · TypeScript · Tailwind · shadcn/ui
Backend	FastAPI · Python · REST API
Computer Vision	OpenCV
OCR	EasyOCR
Field Extraction	Regex
Compliance Engine	Python · Rule-based validation · Compliance checks
Database	PostgreSQL, Prisma, SQLAlchemy 2.0
Reports	ReportLab
Security	JWT · Next Auth.js

IMPLEMENTATION METHODOLOGY - 6 PHASES

01

Foundation

Repo, Next.js + FastAPI skeleton, PostgreSQL schema, auth & API contracts

02

AI EXTRACTION PIPELINE

Image upload, OpenCV preprocessing, OCR, bounding boxes and required field extraction.

03

COMPLIANCE ENGINE

Rule repository, declaration checks, violation classification, severity and confidence handling.

04

EVIDENCE & REPORTING

Store original images and extracted evidence, generate compliance results and PDF reports.

05

PRODUCTIZATION

Dashboard, inspection history, report management, RBAC and audit logging.

06

VALIDATION

Accuracy testing, edge-case testing, end-to-end testing, security checks.

FEASIBILITY OF THE IDEA

TECHNICAL FEASIBILITY

- Built using mature technologies: Next.js, FastAPI, OpenCV, PaddleOCR and PostgreSQL.
- No custom AI model training is required; the core approach combines OCR/CV extraction with deterministic rule-based validation.
- Required declarations can be detected and validated using text, bounding boxes and measurable visual properties.

OPERATIONAL FEASIBILITY

- Follows a simple officer workflow: Scan → Extract → Validate → Review → Report.
- Web-based interface supports product image capture/upload and centralized inspection management.
- NEEDS REVIEW cases allow officers to manually verify uncertain or borderline results.

ECONOMIC & LEGAL FEASIBILITY

- Uses widely available open-source technologies, reducing software licensing dependency.
- Can support centralized deployment without requiring specialized field hardware.
- Final compliance action remains with the authorized officer, while VALIDRA provides evidence and validation results.
- Inspection records, original images and generated reports provide traceability and accountability.

CHALLENGES & MITIGATION STRATEGIES

! OCR errors on real-world packaging

RISK - Glare, curved surfaces, low contrast and complex layouts can reduce OCR accuracy.

MITIGATION — OpenCV preprocessing, perspective correction and OCR confidence scoring; low-confidence extraction is routed to NEEDS REVIEW.

! Font-size measurement from images

RISK — Legal requirements specify physical font sizes, while images provide measurements in pixels.

MITIGATION — Use image calibration/reference measurements to estimate physical size and flag borderline measurements for manual review.

! Different product categories & rule conditions

RISK — Compliance requirements can vary by product category and declaration type.

MITIGATION — Organize compliance checks as configurable rules mapped to the relevant product category and declaration.

! Incorrect or incomplete AI extraction

RISK — OCR may extract incorrect values or miss required declarations.

MITIGATION — Combine OCR confidence + field validation + rule checks and use NEEDS REVIEW when confidence is insufficient.

TARGET AUDIENCE & IMPACT

LEGAL METROLOGY INSPECTORS

- Faster inspection — Scan a product and automatically extract required declarations.
- Easy compliance checking — Rule engine checks MRP, quantity, dates, manufacturer details, font size and placement.
- Evidence-based reporting — Get clear COMPLIANT / NON-COMPLIANT / NEEDS REVIEW results with supporting image evidence.

CONSUMERS

- better product information — Clear and correct MRP, quantity, dates and other declarations help consumers make informed purchases.
- Greater product confidence — Regular compliance checks encourage accurate and readable information on packaged goods.
- Better protection — Faster detection of packaging violations helps reduce misleading or incorrect declarations.

INDUSTRY & E-COMMERCE

- Early compliance checks — Manufacturers can identify label issues before products reach the market.
- Better production quality — Clear compliance requirements encourage manufacturers to maintain accurate packaging and product information.
- Fewer compliance issues — Early detection can help reduce rework, rejected packaging and regulatory violations.

BENEFIT ANALYSIS

SOCIAL BENEFITS

- Better consumer information — Accurate and readable MRP, quantity and product declarations.
- Greater consumer confidence — Consistent compliance checks support trustworthy packaged products.
- Stronger consumer protection — Faster identification of incorrect or missing declarations

ECONOMIC BENEFITS

- Less inspection effort — Automates routine label checking and reduces manual work for officers.
- Higher inspection capacity — Officers can review more products in the same working time.
- Reduced rework — Early identification of packaging errors helps businesses correct issues before release.

ENVIRONMENTAL & SYSTEMIC BENEFITS

- Paperless inspections — Digital evidence and reports reduce dependence on physical records.
- Fewer unnecessary reprints — Early detection helps avoid printing packaging with compliance errors.
- Better compliance insights — Digital inspection data can reveal recurring violations and common problem areas.

EXPECTED IMPACT INDICATORS

FASTER

Reduced manual label-checking effort

MORE COVERAGE

More products inspected per officer

CONSISTENT

Same rule-based checks across inspections

TRACEABLE

Digital evidence preserved with every inspection

ACTIONABLE

Clear COMPLIANT / NON-COMPLIANT / NEEDS REVIEW results

STATUTE & OFFICIAL SOURCES

- Legal Metrology Act, 2009 — Department of Consumer Affairs
- Legal Metrology (Packaged Commodities) Rules, 2011 — as amended
- Department of Consumer Affairs — Legal Metrology guidelines, notifications & amendments
- SIH 2026 — Problem Statement 26034

OCR & IMAGE PROCESSING

- OCR.space API — image-to-text extraction and OCR result overlays
- OpenCV — image preprocessing, resizing and perspective correction
- OCR confidence and extracted text used as inputs for downstream validation

AI / COMPLIANCE PROCESSING

- Groq API — fast LLM inference for structured AI processing
- Rule-based validation — deterministic checks for mandatory declarations
- Confidence-aware validation — uncertain extraction/validation routed to NEEDS REVIEW
- AI output is treated as decision support, not an independent legal verdict

PROJECT REPOSITORY & DOCUMENTATION

- VALIDRA — source code, architecture and technical documentation (github.com/dhruvpatel16120/Validra)
- Next.js — frontend framework
- FastAPI — backend API framework
- PostgreSQL — inspection and compliance data
- ReportLab — compliance report generation

KEY RESEARCH TAKEAWAYS

- OCR enables automated extraction of mandatory declarations from product-label images.
- Visual information matters — Text location and bounding boxes help validate placement and other visual requirements.
- Confidence-based review is essential — Uncertain OCR or extraction results should be flagged for human verification.
- Rule-based validation improves consistency — Compliance checks can be applied systematically across product inspections.

VALIDATION & BENCHMARKING

- Curated product-label images across selected categories
- Ground-truth annotations for MRP, net quantity, dates, manufacturer/packer and consumer-care fields
- Metrics: extraction accuracy, rule-validation accuracy and processing time
- Edge cases: glare, low contrast, curved surfaces and complex layouts

COMPLIANCE PRACTICE

- Mandatory declarations defined under the Packaged Commodities Rules
- Validation of presence, value, format and applicable visual requirements
- Evidence preservation through original images and extracted data
- RBAC and audit logs for inspection traceability